

Nikhil Sudhan J

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EDUCATION

B.Tech – Computer Science and Engineering

Kalasalingam Academy of Research and Education

2022 – 2026

EXPERIENCE

Cloud & DevOps Intern - Wipro, Bangalore

May- June 2023

- Containerized and deployed cloud-native applications using **Docker** and **Kubernetes** on **Google Cloud Platform**.
- Built **GitOps-based CI/CD pipelines** enabling automated deployment and version control.

Software Engineer Intern - Sonata Software, Chennai

May – June 2024

- Built a **Flask** app integrated with **Google Cloud Document AI** for automated document extraction.
- Implemented **OAuth2 authentication** and deployed on a Windows Server VM.

PROJECTS

Pluto – AI Integrated UAV Command System

Python, Flask, MAVLink, React, REST APIs

- Built a full-stack UAV platform: React frontend for mission authoring, Flask backend for prompt-to-instruction translation, and MAVLink layer for live drone telemetry and control.
- Engineered a modular command engine exposing REST APIs for real-time mission execution with dynamic re-routing and structured error handling across GPS-denied flight scenarios.

Real-Time 3D Object Tracking on Edge AI

PyTorch, TensorRT, KITTI, Jetson Nano, CUDA

- Designed a lightweight monocular 3D perception model using depthwise separable convolutions; optimized with TensorRT achieving **20+ FPS** on Jetson Nano at sub-1GB VRAM.
- Benchmarked against full-precision baselines validating production-ready accuracy-latency trade-offs with zero cloud dependency on constrained edge hardware.

Adaptive Vector Retrieval Engine

Python, FAISS, SQLite, REST APIs

- Architected a cold-hot vector storage system using FAISS with on-demand embedding loading; exposed REST APIs for ingestion, search, and lifecycle management.
- Benchmarked retrieval latency and scalability vs. full in-memory vector DBs, demonstrating improved scalability with minimal latency trade-offs under concurrent query spikes.

TECHNICAL SKILLS

Languages: Python, C, Dart, SQL

Backend & APIs: Flask, REST API Design, Microservices, OAuth2, WebSockets

AI / Vision: YOLO, LoRA, FAISS, Vector Search

Robotics / UAV: ROS2, MAVLink, Gazebo, Jetson Nano, Edge Deployment

Cloud / DevOps: Docker, Kubernetes, GitOps, CI/CD, GCP, Supabase

Tools: Linux, Git, Fusion360, Wireshark, Burp Suite, Metasploit, Cursor, Claude, Codex

CERTIFICATIONS

Oracle - Oracle Cloud Infrastructure AI Foundations Associate (2025)

Udemy - Machine Learning: NLP

Cisco - Introduction to Networks